

UniKL PJ Research Collaboration Portal - Manual

Setup & Hardening Guide

This document lists every command and configuration edit required to set up your environment manually on your Ubuntu VM. You can copy these commands one-by-one, run them, and capture screenshots for your JURUS project report.

■■ Step 1: Install System Packages

Run this command to update your system repositories and install Nginx (web server), PostgreSQL (database), Fail2ban (intrusion prevention), UFW (firewall), SSH daemon, and PAM/security components:

```
sudo apt-get update
sudo apt-get install -y python3-pip python3-venv nginx postgresql postgresql-contrib libmagic1 fail2ban ufw
```

Screenshot opportunity: Capture the successful installation logs showing packages being installed.

■ Step 2: Configure Directory Permissions

Create secure, isolated directories for document storage, custom logging, and static assets, assigning ownership to your administrative user (`jurus`) and web server group (`www-data`):

```
# Create directories
sudo mkdir -p /var/www/uniklpj_uploads
sudo mkdir -p /var/log/uniklpj
sudo mkdir -p /var/www/uniklpj/static

# Assign permissions to upload directory
sudo chown -R jurus:www-data /var/www/uniklpj_uploads
sudo chmod 770 /var/www/uniklpj_uploads

# Assign permissions to application logs directory
sudo chown -R jurus:www-data /var/log/uniklpj
sudo chmod 770 /var/log/uniklpj
```

```
# Copy static assets to secure web directory and configure permissions
sudo cp -r /home/jurus/Documents/Project/UniPortal-Jurus/uniklpj_portal/static/* /var/www/uniklpj/
sudo chown -R jurus:www-data /var/www/uniklpj/static
sudo chmod -R 750 /var/www/uniklpj/static
```

Verify upload permissions using `ls -ld /var/www/uniklpj_uploads` (should output `drwxrwx--- 2 jurus www-data`).

■ Step 3: Python Environment Setup

Isolate the web application's dependencies in a Python Virtual Environment (venv):

```
cd /home/jurus/Documents/Project/UniPortal-Jurus/

# Create the virtual environment
python3 -m venv venv

# Upgrade pip inside venv
./venv/bin/pip install --upgrade pip

# Install requirements
./venv/bin/pip install -r requirements.txt

# Ensure file ownership is clean
sudo chown -R jurus:jurus /home/jurus/Documents/Project/UniPortal-Jurus/
```

■ Step 4: Generate SSL Certificate (TLS 1.3)

Generate a self-signed OpenSSL certificate for the portal (valid for 365 days) and secure the private key:

```
# Generate the keys
sudo openssl req -x509 -nodes -days 365 -newkey rsa:2048 \
  -keyout /etc/ssl/private/uniklpj.key \
  -out /etc/ssl/certs/uniklpj.crt \
  -subj "/C=MY/ST=Selangor/L=Petaling Jaya/O=UniKL PJ/OU=Cybersecurity/CN=localhost"

# Hardened file permissions
sudo chmod 600 /etc/ssl/private/uniklpj.key
```

```
sudo chmod 644 /etc/ssl/certs/uniklpj.crt
```

■ Step 5: SSH Daemon Hardening

We will modify `/etc/ssh/sshd_config` to enforce keys only and change the default port.

1. Create a backup first: `bash sudo cp /etc/ssh/sshd_config /etc/ssh/sshd_config.bak`
2. Open the file in a terminal editor: `bash sudo nano /etc/ssh/sshd_config`
3. Locate or add the following directives (make sure they are not commented out with a #):
`text`
`Port 2222 PermitRootLogin no PasswordAuthentication no PubkeyAuthentication yes`
`MaxAuthTries 3 X11Forwarding no`
4. Save and exit (in Nano: press `Ctrl+O`, then `Enter`, then `Ctrl+X`).

Restart the SSH daemon to apply: `bash sudo systemctl restart ssh`

Warning: Ensure you have copied your client SSH public key (`authorized_keys`) into `/home/jurus/.ssh/` before logging out of this terminal, otherwise password access will be rejected.

■ Step 6: Configure UFW Firewall

Enforce a default-deny rule and open only the necessary ports:

```
# Reset UFW to default settings
sudo ufw --force reset

# Set default policies
sudo ufw default deny incoming
sudo ufw default allow outgoing

# Allow specific secure ports
sudo ufw allow 2222/tcp
sudo ufw allow 80/tcp
sudo ufw allow 443/tcp

# Enable firewall
sudo ufw --force enable
```

Verify status using `sudo ufw status verbose`.

■ Step 7: Configure PAM Password Complexity

Enforce complexity policies for local system accounts:

1. Open the PAM common-password file: `bash sudo nano /etc/pam.d/common-password`
 2. Locate the line containing `pam_pwquality.so` (usually around line 23): `text password requisite pam_pwquality.so retry=3`
 3. Edit that line to include the complexity requirements: `text password requisite pam_pwquality.so retry=3 minlen=8 ucredit=-1 lcredit=-1 dcredit=-1 ocredit=-1`
(This enforces: minimum 8 characters, at least 1 uppercase, 1 lowercase, 1 digit, and 1 special character).
 4. Save and close.
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■■ Step 8: Setup PostgreSQL Database

Configure your database and establish user permissions.

1. Ensure the PostgreSQL service is active: `bash sudo systemctl start postgresql sudo systemctl enable postgresql`
2. Switch to the `postgres` administrator account and launch the SQL terminal: `bash sudo -u postgres psql`
3. Execute these SQL statements inside the `psql` terminal one-by-one: ```sql -- Create database CREATE DATABASE uniklpj_db;`

```
-- Create portal user with secure password CREATE USER portal_user WITH ENCRYPTED  
PASSWORD 'PortalSecure@2026!';
```

```
-- Grant privileges GRANT ALL PRIVILEGES ON DATABASE uniklpj_db TO portal_user;
```

```
-- Exit the SQL terminal \q 4. Grant schema permissions to the user inside the  
database: bash sudo -u postgres psql -d uniklpj_db -c "GRANT ALL ON SCHEMA public TO  
portal_user;" ````
```

■ Step 9: Configure Nginx & Systemd Services

A. Deploy Systemd Background Service

1. Copy the systemd configuration file: `bash sudo cp /home/jurus/Documents/Project/UniPortal-Jurus/uniklpj.service /etc/systemd/system/uniklpj.service`
2. Reload daemon, enable service, and start Unicorn: `bash sudo systemctl daemon-reload sudo systemctl enable uniklpj sudo systemctl start uniklpj` *Check status using `sudo systemctl status uniklpj`.*

B. Configure Nginx Reverse Proxy

1. Copy the Nginx configuration: `bash sudo cp /home/jurus/Documents/Project/UniPortal-Jurus/nginx_uniklpj.conf /etc/nginx/sites-available/uniklpj`
2. Enable the site configuration: `bash sudo ln -sf /etc/nginx/sites-available/uniklpj /etc/nginx/sites-enabled/`
3. Remove Nginx default index site: `bash sudo rm -f /etc/nginx/sites-enabled/default`
4. Verify config and reload Nginx: `bash sudo nginx -t sudo systemctl restart nginx` *Open `https://localhost` in the browser to confirm the dashboard renders securely.*

■ Step 10: Configure Fail2ban Jails

Enable Fail2ban active defense.

1. Create a custom jail file: `bash sudo nano /etc/fail2ban/jail.d/uniklpj.local`
2. Paste this configuration: ```ini [sshd] enabled = true port = 2222 filter = sshd logpath = /var/log/auth.log maxretry = 5 bantime = 1800 findtime = 600`

`[nginx-http-auth] enabled = true filter = nginx-http-auth port = http,https logpath = /var/log/nginx/error.log maxretry = 5 bantime = 1800 findtime = 600` 3. Save, close, and restart Fail2ban: `bash sudo systemctl restart fail2ban sudo systemctl enable fail2ban` ``` *Check active jails with sudo fail2ban-client status`.*`